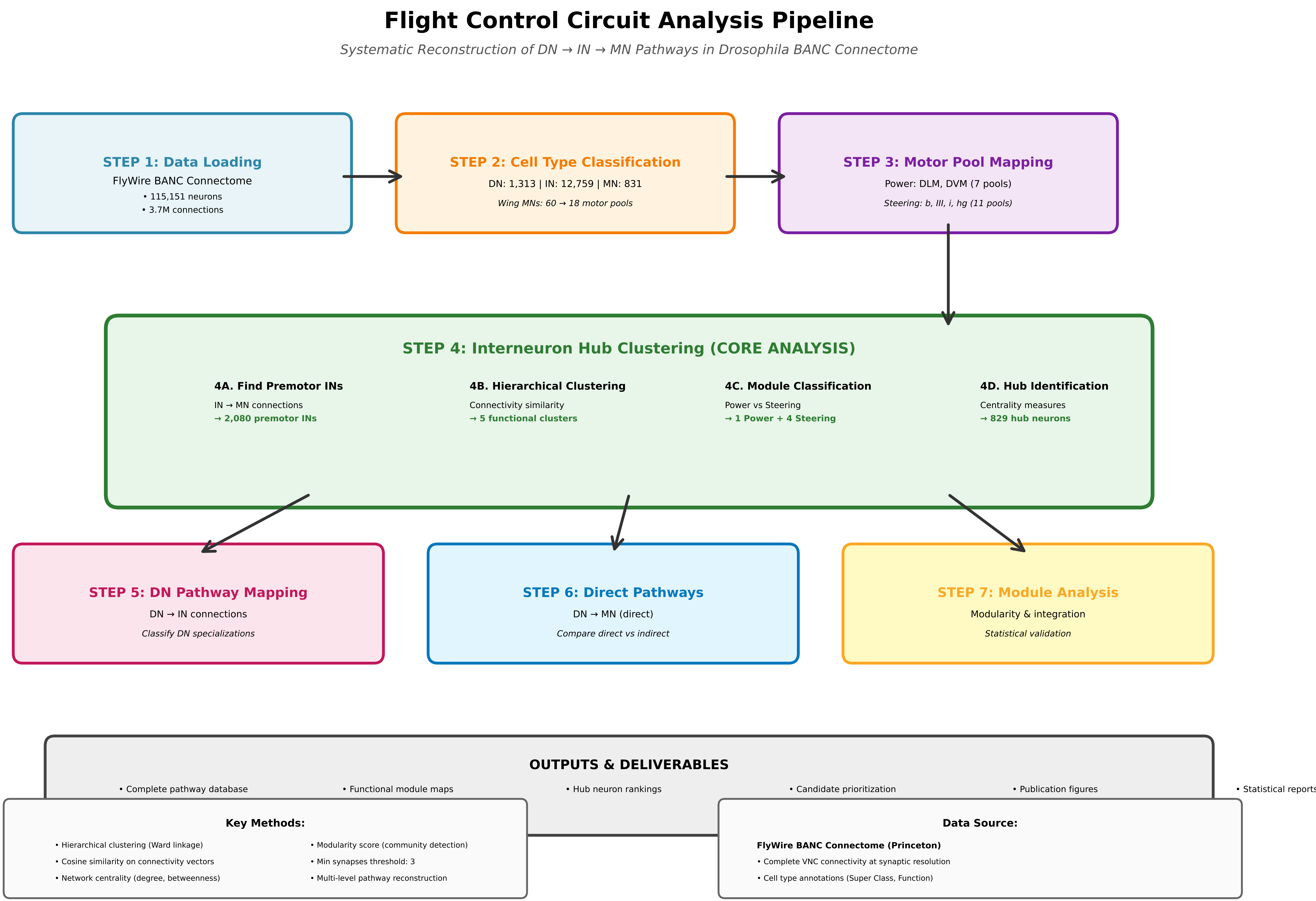
Flight Control Circuit Analysis Pipeline
Systematic Reconstruction of DN → IN → MN Pathways in Drosophila BANC Connectome
STEP 1: Data Loading
FlyWire BANC Connectome
• 115,151 neurons
• 3.7M connections
STEP 2: Cell Type Classification
DN: 1,313 | IN: 12,759 | MN: 831
Wing MNs: 60 → 18 motor pools
STEP 3: Motor Pool Mapping
Power: DLM, DVM (7 pools)
Steering: b, III, i, hg (11 pools)
STEP 4: Interneuron Hub Clustering (CORE ANALYSIS)
4A. Find Premotor INs
IN → MN connections
→ 2,080 premotor INs
4B. Hierarchical Clustering
Connectivity similarity
→ 5 functional clusters
4C. Module Classification
Power vs Steering
→ 1 Power + 4 Steering
4D. Hub Identification
Centrality measures
→ 829 hub neurons
STEP 5: DN Pathway Mapping
DN → IN connections
Classify DN specializations
STEP 6: Direct Pathways
DN → MN (direct)
Compare direct vs indirect
STEP 7: Module Analysis
Modularity & integration
Statistical validation
OUTPUTS & DELIVERABLES
• Complete pathway database
• Functional module maps
• Hub neuron rankings
• Candidate prioritization
• Publication figures
• Statistical reports
Key Methods:
• Hierarchical clustering (Ward linkage)
• Cosine similarity on connectivity vectors
• Network centrality (degree, betweenness)
• Modularity score (community detection)
• Min synapses threshold: 3
• Multi-level pathway reconstruction
Data Source:
FlyWire BANC Connectome (Princeton)
• Complete VNC connectivity at synaptic resolution
• Cell type annotations (Super Class, Function)